

Optimized Quantum Machine Learning Pulse Sequences: A Finite Mathematical Framework for Neuromorphic Imaging

Cartik Sharma, Neuromorph System

Abstract: Recent advancements in quantum algorithm-driven radiofrequency pulse sequences necessitate robust theoretical foundations. Here we present a set of finite mathematical formulations to formalize the most recent dual spin-echo generated waveforms. We utilize discrete spatial Laplacians for inflection-point detection and Euler integration methods. Crucially, we investigate the amplitude modulation observed in the last two pulse sequences synthesized in our environment, establishing a definitive discrete topological framework for neurovascular imaging architectures.

1. Finite Mathematical Analysis of the Last Two Pulse Sequences

The latest phase-amplitude arrays reflect a stable progression of optimized RF transitions. Utilizing the discrete time formulation, the RF envelope B1 is modeled over N finite steps:

$$B_{1,k+1} = B_{1,k} + \alpha \cdot \cos(\omega \cdot t_k) \cdot \Delta t$$

where α is the quantum control scalar. This formulation is explicitly applied to the last two pulse variations (custom_sequence and custom_sequenceA) recently executed, indicating stable performance limits.

2. Finite Difference Laplacian Operators

To compute optimal readout intervals and structural gradients during the sequence execution, the spatial Laplacian is approximated finitely:

$$\nabla^2 M_{i,j} \approx [M_{i+1,j} + M_{i-1,j} + M_{i,j+1} + M_{i,j-1} - 4M_{i,j}] / h^2$$

where M represents the discrete transverse magnetization. This prevents the computational instability and singularities characteristic of continuum gradient-descent.

3. Diffeomorphic Mappings using Discrete Euler Integration

The spatial mapping ϕ of the pulse responses maintains topological tissue integrity, ensuring no physical folding occurs. A stationary finite velocity field $v(x)$ drives the diffeomorphism:

$$\phi_{k+1}(x) = \phi_k(x) + (1/N) \cdot v(\phi_k(x))$$

The positive determinant Jacobian generated strictly tracks structural deformation induced by the latest temporal TR/TE settings.